



NAVISense

April 2026
NAVISense Team

DEVELOPMENT

DRAFT

V1.0

Prototype Development Report

A focused technical note on progress, decisions, risks, and next steps.

Executive signal

Use this cover note to frame the report in one sentence: what changed, why it matters, and what the reader should take away before reading the details.

1 Executive Summary

This report summarizes the current state of the NAVISense Vest workstream. Replace this paragraph with a concise overview of what happened, what changed, and why the update matters for the project.

62%**PROGRESS**

Prototype integration is moving through the active build phase.

2**KEY RISKS**

Power budget and sensor latency require continued validation.

1**NEXT FOCUS**

Prepare reliable testing sessions with repeatable scenarios.

2 Context

Explain the background of this report: what triggered the work, what decisions had already been made, and which part of the system this update covers.

Useful framing

Keep this section short. A report should not repeat the whole project proposal; it should make the current update easy to understand.

3 Work Completed

- Replaced this bullet with a concrete task or decision.
- Mention the responsible person or subsystem when useful.
- Link each completed item to evidence, measurements, screenshots, or diagrams.

4 Technical Notes

4.1 Architecture

Use this subsection for diagrams, component interactions, database/API decisions, firmware notes, or implementation details.

Example note: The vest-to-application communication path should describe sensor input, embedded processing, data transmission, dashboard storage, and user feedback.

4.2 Evidence

Add measurements, test data, screenshots, or references here.

Drop an image, chart, circuit capture, or dashboard screenshot here.

5 Risks and Mitigations

Risk	Impact	Mitigation
Sensor latency	Navigation feedback may feel late or confusing.	Run controlled latency tests and define acceptable thresholds.
Power draw	Battery life may be lower than expected.	Profile current draw by subsystem and prioritize sleep states.
Mechanical comfort	The vest may become uncomfortable over longer use.	Collect user feedback during short fitting sessions.

6 Timeline

COMPLETED

Week 01

Initial research, component selection, and report framing.

IN PROGRESS

Week 02

Prototype integration and early validation.

NEXT

Week 03

Structured testing, evidence collection, and blog publication.



7 Next Steps

- Define the next measurable deliverable.
- Assign owners and deadlines.
- Decide what evidence should be collected for the next full report.

8 Appendix

Use the appendix for raw notes, tables, screenshots, links, or references that are useful but too detailed for the main narrative.